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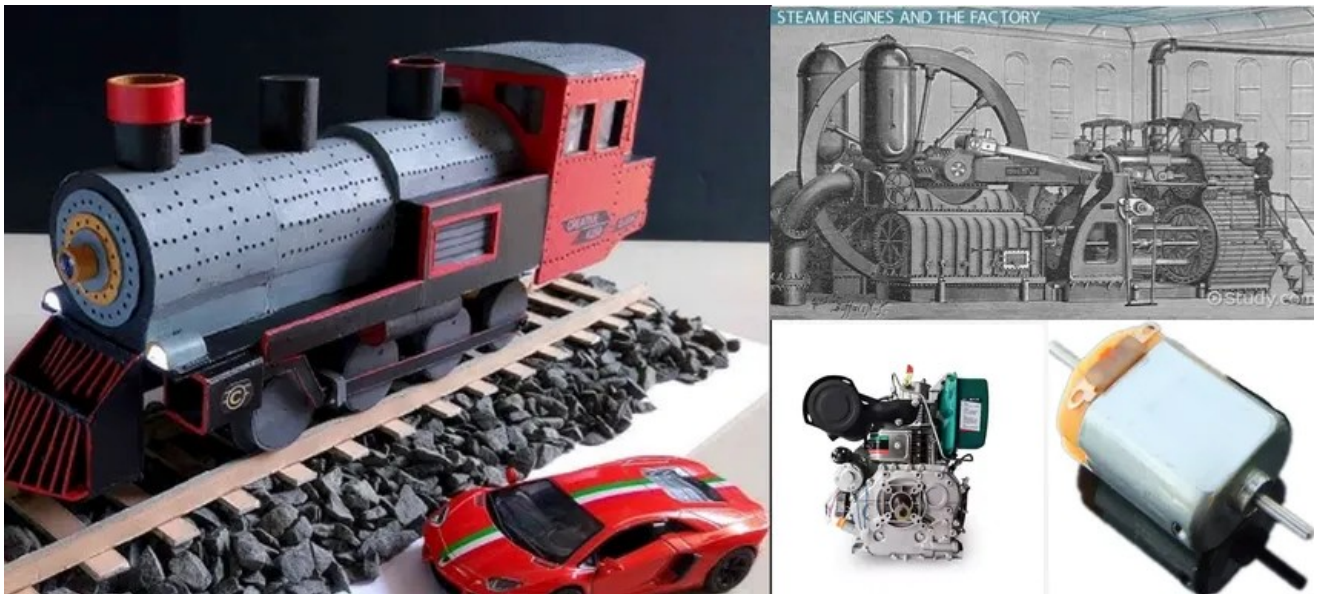
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# Model Compression: Pruning, Quantization, Distillation and Binarization

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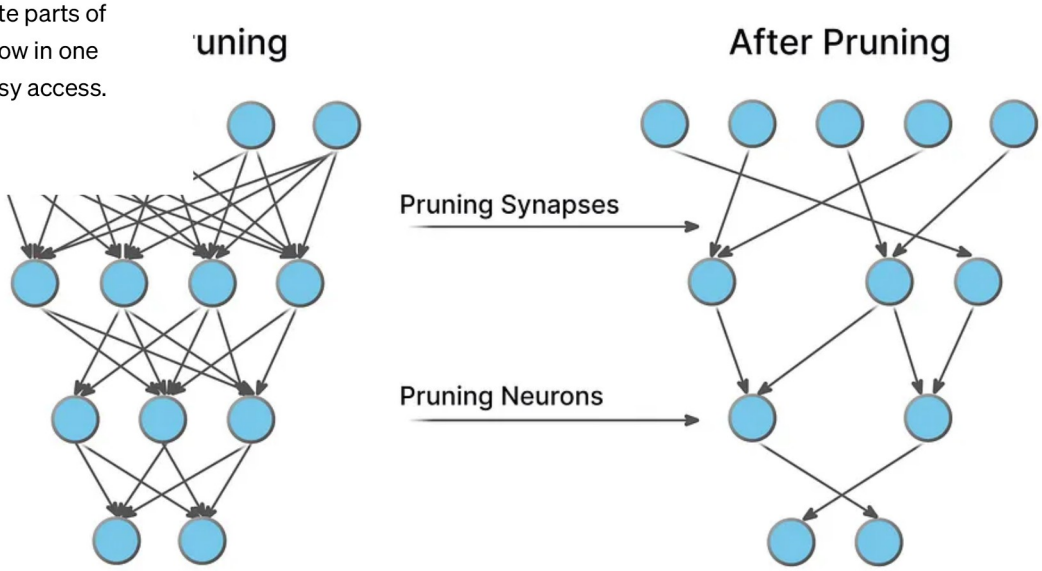

From Steam Engines to Electric Motors

In short, Pruning is removing unnecessary parts of the model. Quantization is simplifying numerical calculations. Distillation is transferring knowledge from a large model to a smaller one, and Binarization is reducing computations to binary operations.

1. **Pruning: Cutting the redundant**
   Imagine you have a huge library with thousands of books, but only a few are actually useful for your daily reading. The rest just take up space and make it harder to find what you need. Similarly, AI models can have a lot of unnecessary parts (called parameters) that don't contribute much to making decisions.

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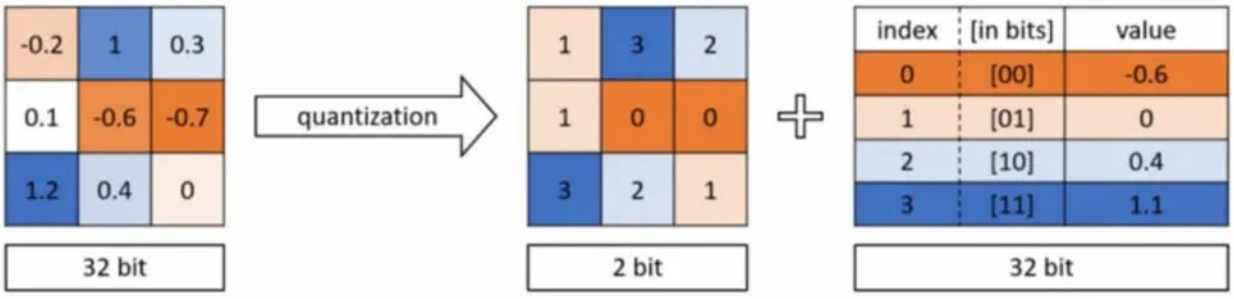



A neural network structure, before and after pruning. Source: Pruning Neural Networks

Pruning is like decluttering your library. It removes these unimportant parts from the model, making it smaller and faster without affecting its performance. For example, if an AI model used to take 10 seconds to recognize a photo but now takes just 5 after pruning.

**2. Quantization: Simplifying the Math**

Think about doing math with big numbers versus small ones. Big numbers require more time, effort, storage and memory. Similarly, AI models use precise calculations (like using decimals) which can be resource-heavy.




Source: 4 Popular Model Compression Techniques Explained

Quantization simplifies these calculations by reducing the precision of the numbers used. Instead of using exact decimal values, it rounds them to smaller, simpler

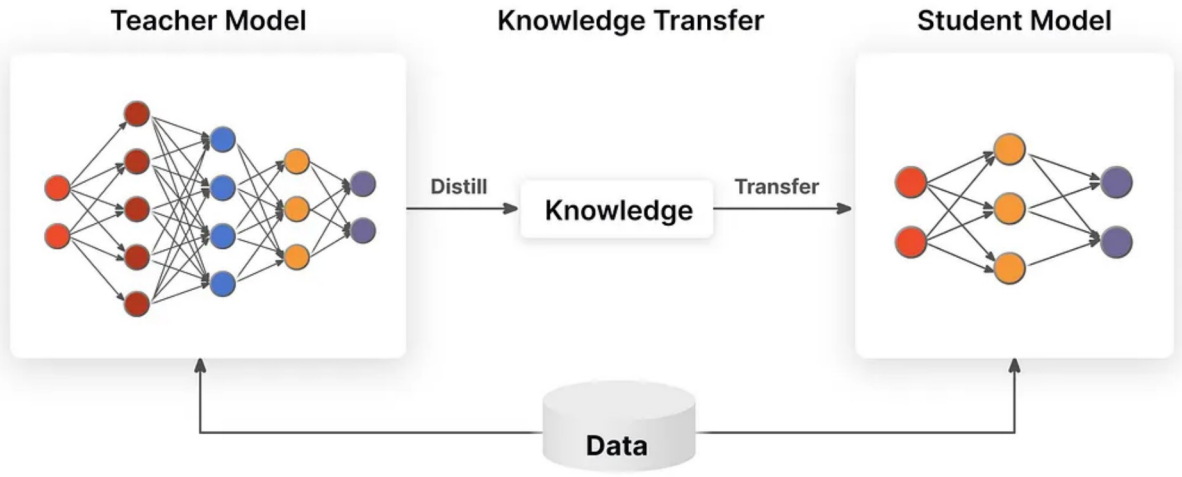
numbers. This makes the model lighter and faster. For instance, if a number was 3.14159, quantization might make it just 3 or even round it up to 3.2.

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lel (the student) learns from a larger, more powerful one (the hares its knowledge with the student, who then becomes skilled ks independently but without the bulk.




An illustration of the knowledge distillation process. Source: Knowledge Distillation: A Survey

For example, imagine teaching a child how to solve a math problem by first explaining it in simple terms and then letting them try it on their own. The result is a smarter, smaller model that can handle the task efficiently.

**4. Binarization: Black and White Thinking**

What if everything was reduced to just two choices — like yes or no? That's the idea behind binarization . Instead of using a wide range of values, AI models are simplified to use only 0s and 1s (binary). This drastic simplification makes the model extremely lightweight and fast but still capable enough for many tasks.

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Source: Deep Compression

Think of it like creating a simple on/off switch for decisions instead of having a complicated dial with hundreds of settings. While it might not handle every task perfectly, it's surprisingly effective for basic operations.

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There also has Low-rank factorization that decompose weight matrices into smaller "low-rank" matrices. So its simplifying matrix operations. Low-Rank Factorization is a technique that leverages the concept of matrix decomposition to compress neural

network layers. The idea is based on the observation that many matrices used in neural networks are approximately low-rank, meaning they can be represented as the product

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ı't just about making things smaller; it's about making AI accessible and efficient everywhere. Model compression techniques like Pruning, Quantization, Distillation, and Binarization are essential tools for keeping AI models efficient and accessible. They help in reducing the size of models without sacrificing their performance, ensuring that we can enjoy faster and smarter technologies every day.

### VS. Fine-Tuning

Fine-tuning involves adjusting or optimizing an existing model to perform better on a specific task or dataset. This process often requires retraining the model with additional data or tweaking its parameters to align more closely with the desired outcome. The goal is to improve accuracy or performance for the target use case. Fine-Tuning could works with start with a pre-trained model, retrain it on your custom dataset and adjust hyperparameters for better results. One possible use case is Fine-tuning a language model to improve sentiment analysis for customer reviews.

| **Aspect** | **Model Compression** | **Fine-Tuning** |
|---|---|---|
| Goal | Make the model smaller, faster, and more efficient. | Improve performance on a specific task or dataset. |
| Techniques Used | Pruning, Quantization, Distillation, Binarization. | Retraining with new data, adjusting hyperparameters, transfer learning. |
| Impact on Size | Reduces model size and complexity. | Model size may remain the same or even increase slightly if more layers are added. |
| Use Case | Deploying models on resource-constrained devices (e.g., mobile phones). | Enhancing performance for a specific application (e.g., improving accuracy in NLP tasks). |
| Performance Trade-off | May slightly reduce accuracy to achieve efficiency gains. | Typically improves accuracy or performance at the cost of potentially increasing model size or training time. |

### VS. Retrieval-Augmented Generation (RAG)

My favorite one. RAG combines the power of large language models with external knowledge sources. Instead of retraining the model, RAG retrieves information from documents or databases and feeds it into the model during generation. This allows the model to access up-to-date information without requiring extensive fine-tuning. Using RAG to answer questions about company policies by pulling from an internal document repository.

I want to use one example: a steam engine — powerful and capable of handling complex tasks with high precision, more powerful. However, electric motors will revolutionize industries by bringing power into every homes and everyday devices.

Happy reading, happy coding!

## References:

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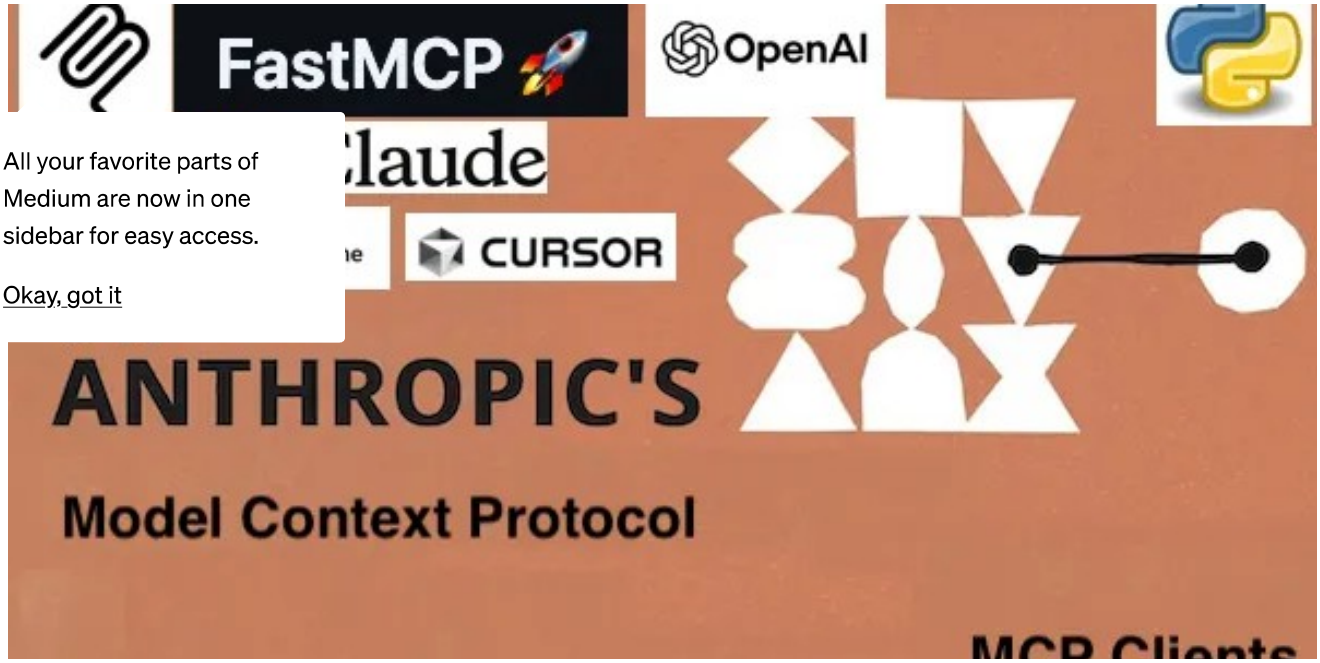
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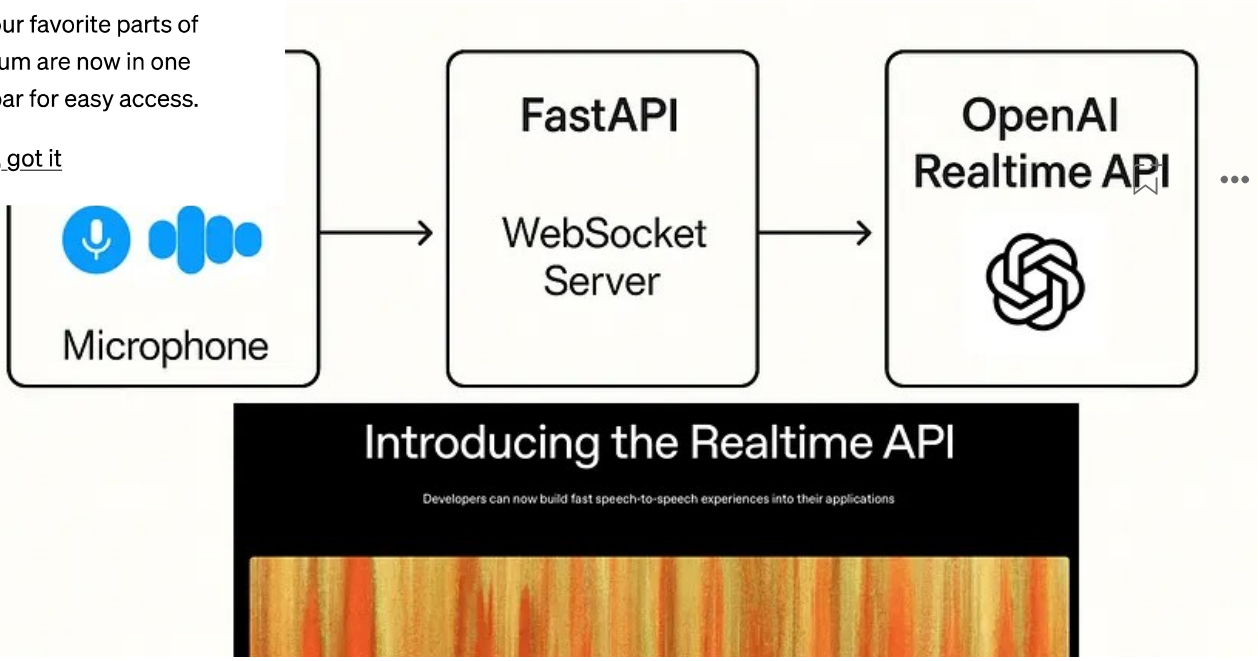


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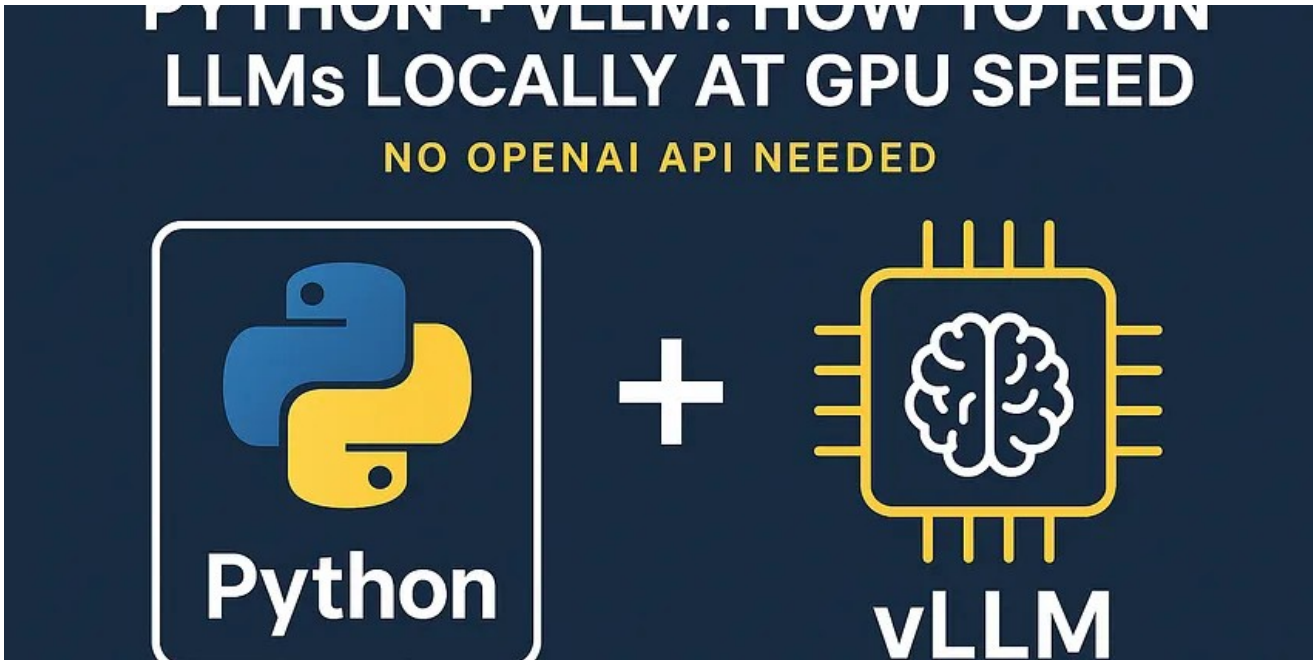
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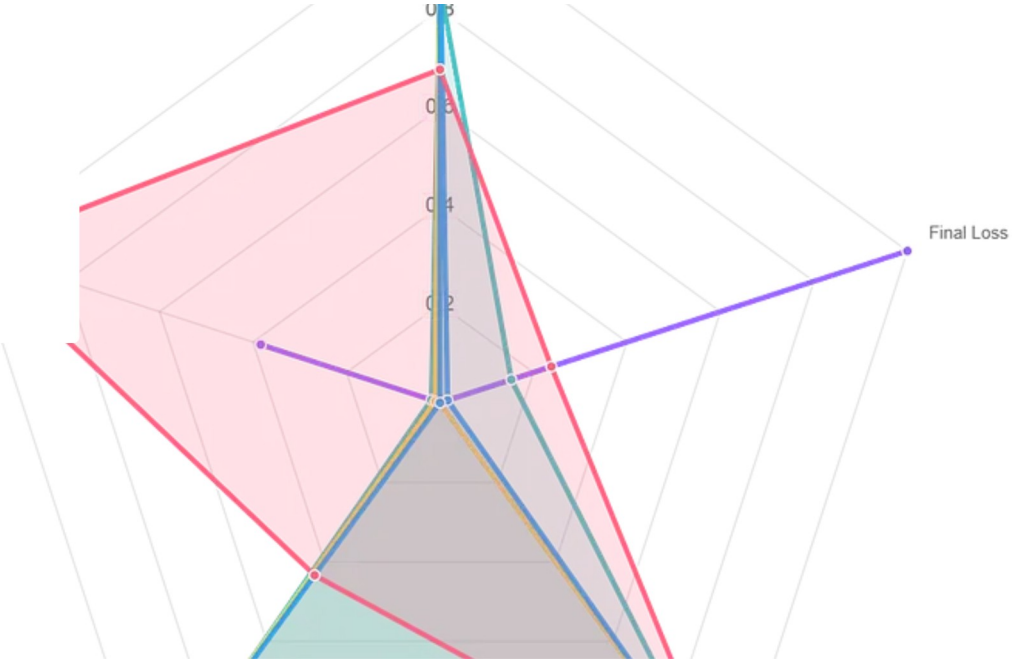
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## P100 vs V100 vs A100 vs H100 for AI Inference: I Ran Them All So You Don't Have To

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ABSTRACT

Post-training alignment often reduces LLM diversity, leading to a phenomenon known as *mode collapse*. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: *typicality bias* in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it on preference datasets empirically, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce ***Verbalized Sampling (VS)***, a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., "Generate 5 jokes about coffee and their corresponding probabilities"). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by 1.6-2.1× over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

**Problem: *Typicality Bias* Causes Mode Collapse**

Tell me a joke about coffee

**Solution: Verbalized Sampling (VS) Mitigates Mode Collapse**

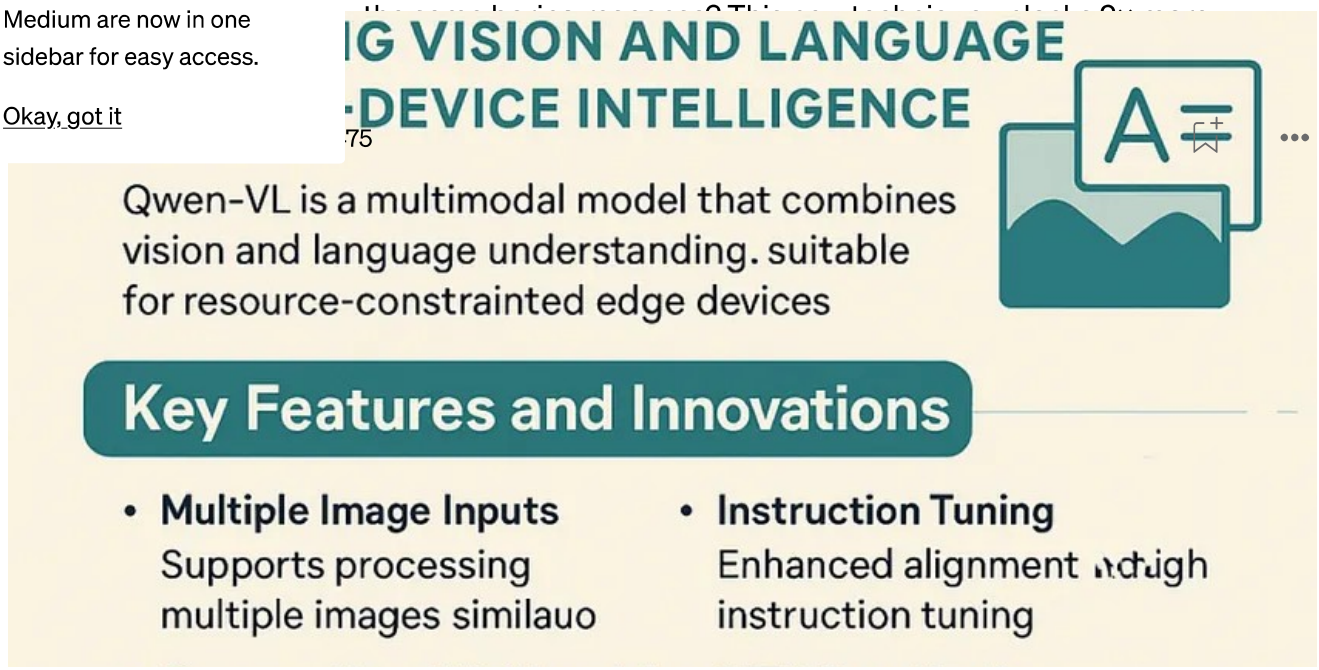
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